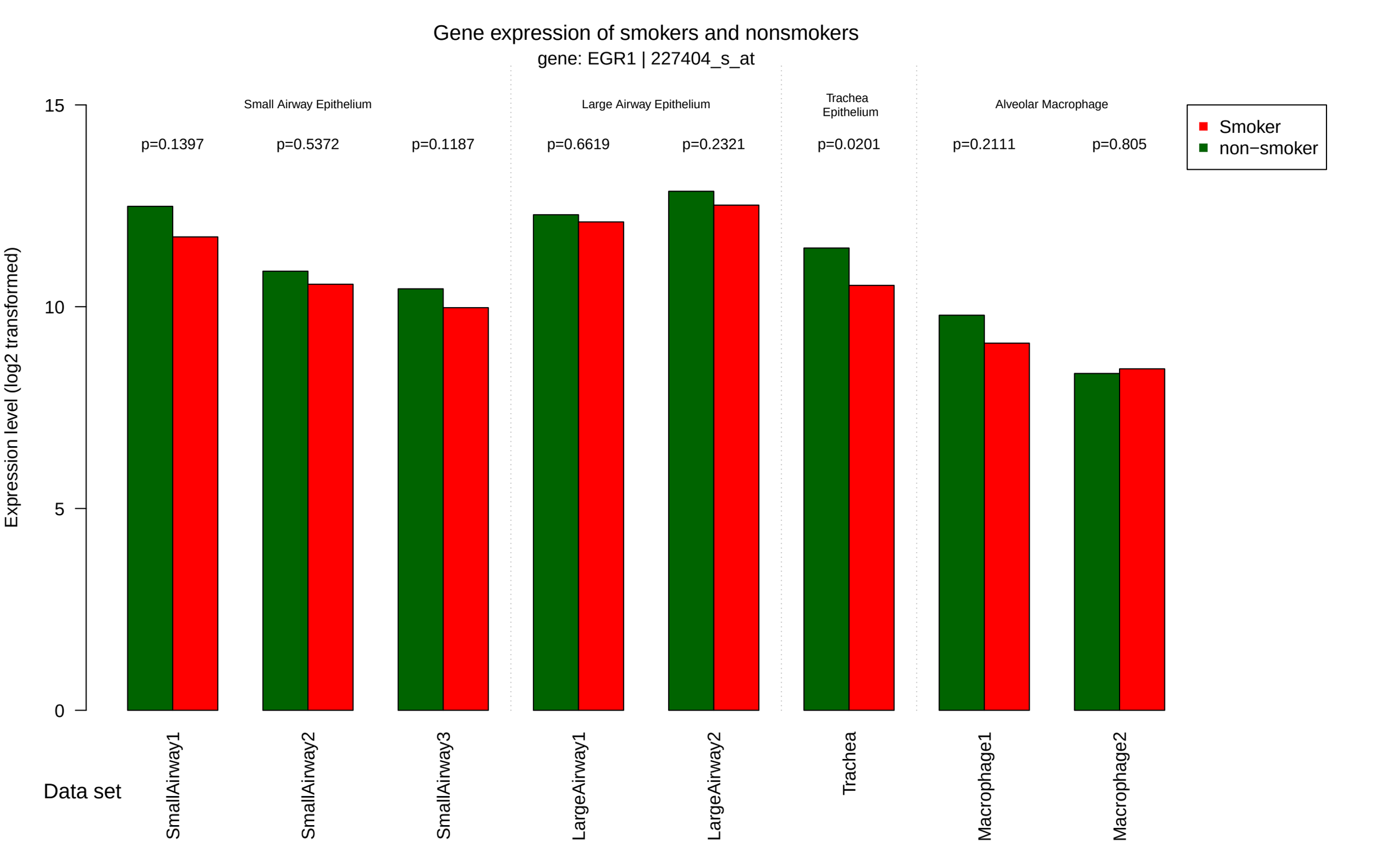
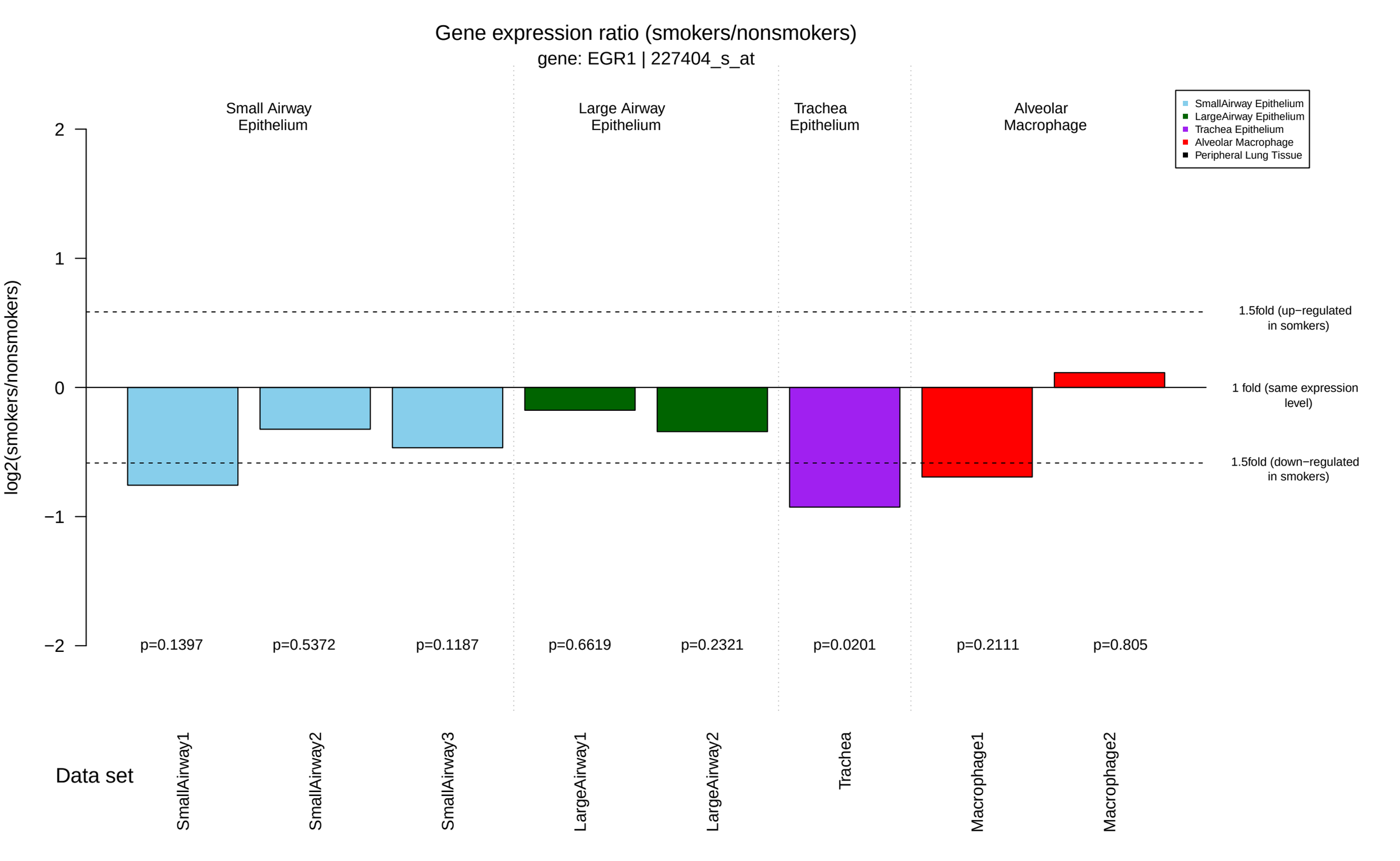
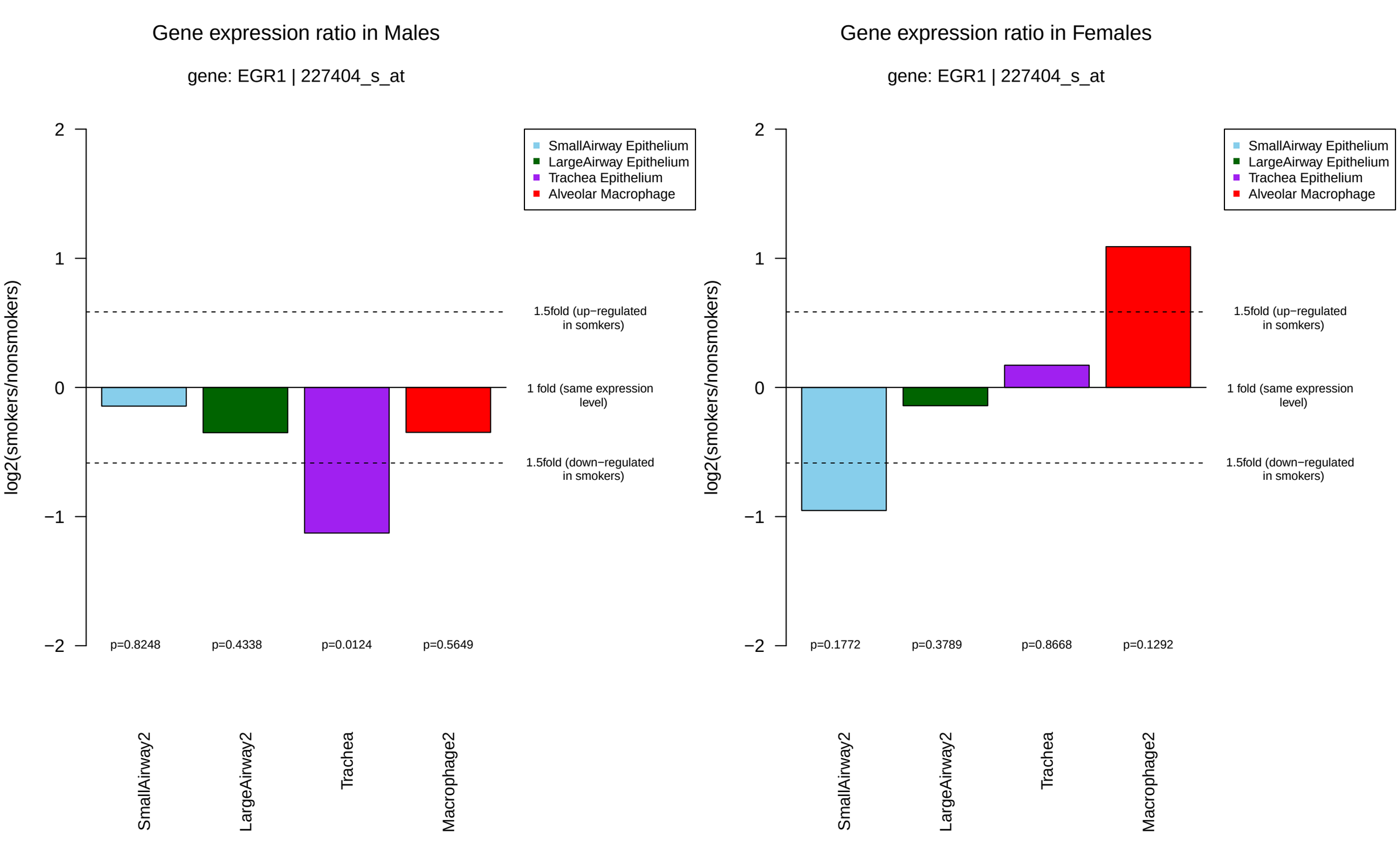
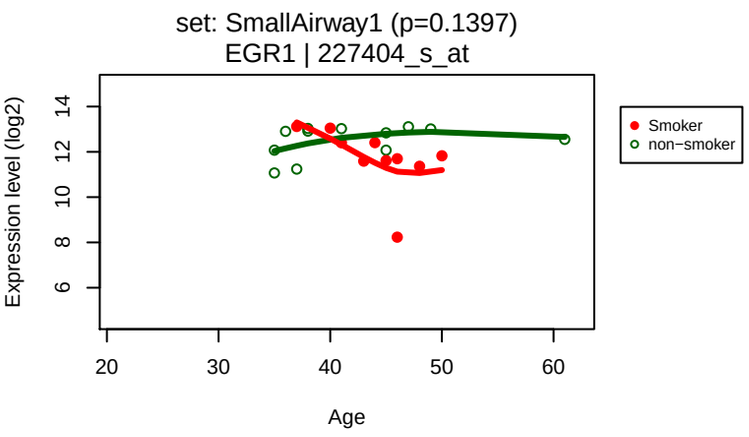
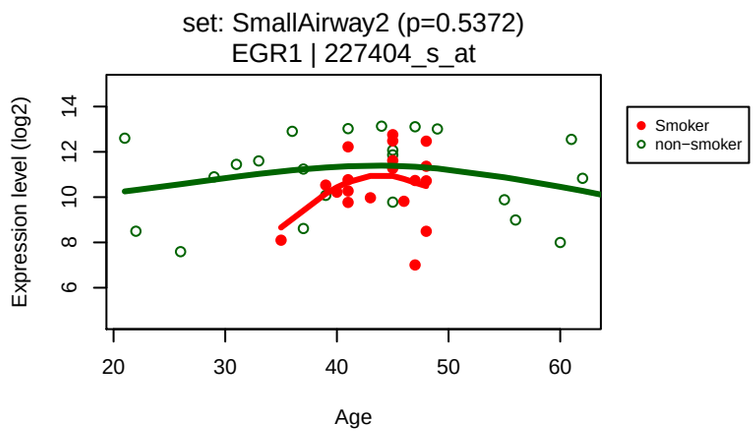
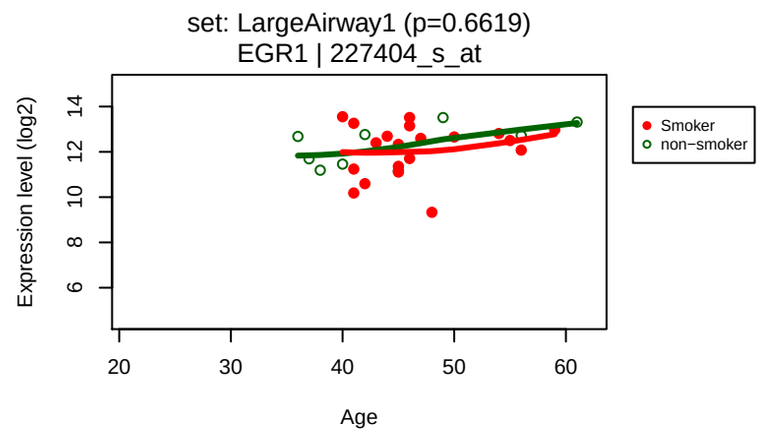

Gene expression of smokers and nonsmokers
gene: EGR1 | 227404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1397
p=0.5372
p=0.1187
p=0.6619
p=0.2321
p=0.0201
p=0.2111
p=0.805
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

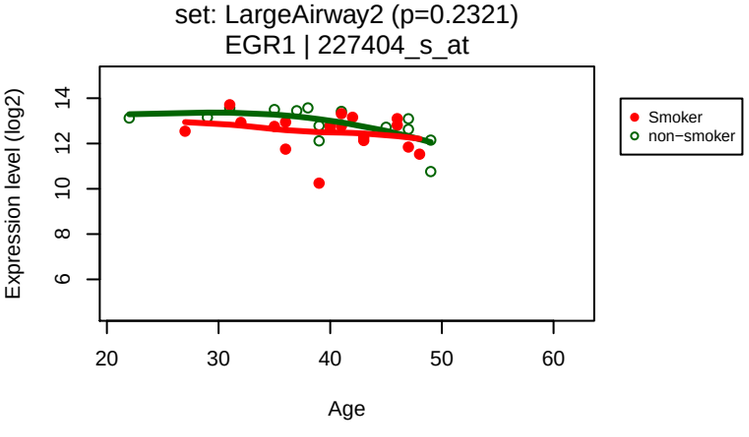

Gene expression ratio (smokers/nonsmokers)
gene: EGR1 | 227404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1397
p=0.5372
p=0.1187
p=0.6619
p=0.2321
p=0.0201
p=0.2111
p=0.805
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

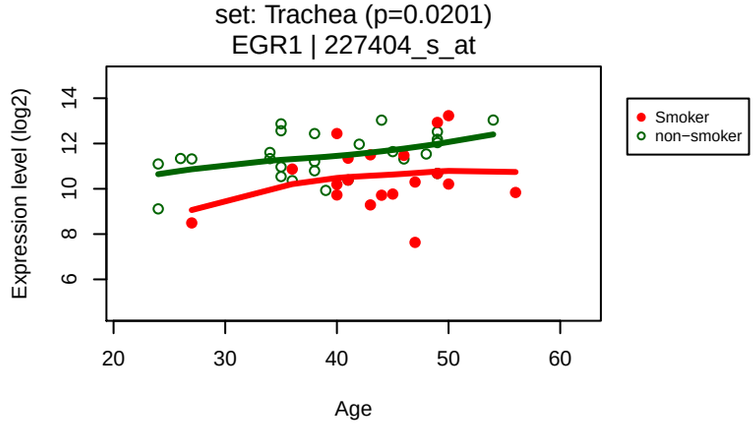

Gene expression ratio in Males
gene: EGR1 | 227404_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8248
p=0.4338
p=0.0124
p=0.5649
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EGR1 | 227404_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1772
p=0.3789
p=0.8668
p=0.1292
SmallAirway2
LargeAirway2
Trachea
Macrophage2

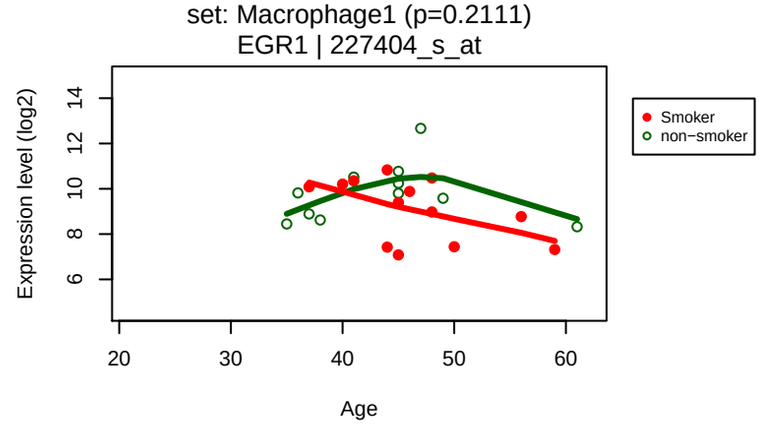

set: SmallAirway1 (p=0.1397)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker

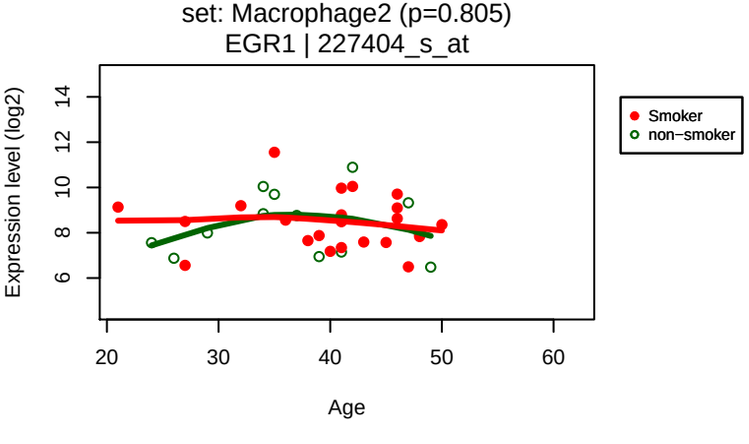

set: SmallAirway2 (p=0.5372)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker

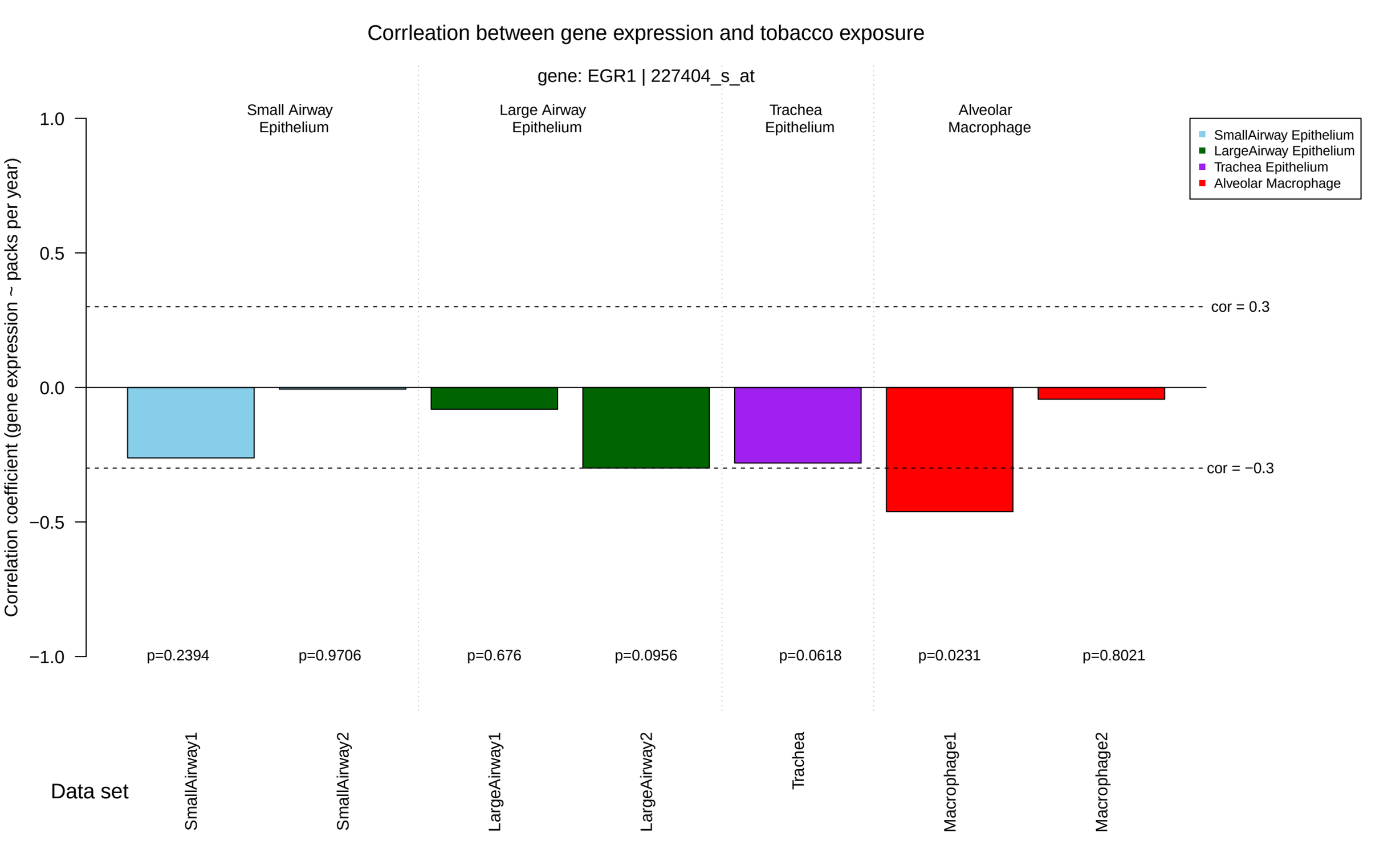

set: LargeAirway1 (p=0.6619)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2321)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0201)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2111)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.805)
EGR1 | 227404_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EGR1 | 227404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2394
p=0.9706
p=0.676
p=0.0956
p=0.0618
p=0.0231
p=0.8021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2